BRANSON'S AND BEN'S GROCERY

COSC 304: Introduction to Databases
Developed by: Branson and Ben (Team 20)
Date: December 3, 2025
GitHub Repository: cosc-304-2025/development-project-lab-7-10-team20

User ID's and Password

- Username: **arnold**, Password: **304Arnold! Admin Account**
- Username: **bobby**, Password: **304Bobby!**
- Username: **candace**, Password: **304Candace!**

ATTENTION!

If you attempt to run this code. You might need to change the jdbc.jsp connection database (change urlForLoadData to the commented out one). You must run loaddata.jsp FIRST, and then run LoadCursedData.jsp for the site to function properly, thanks!

EXECUTIVE SUMMARY

Branson’s and Ben’s Grocery is an online grocery platform designed to give customers a fast, convenient, and reliable way to shop for everyday essentials. Our store provides a seamless digital experience where users can browse products, manage their accounts, and place secure orders—all from the comfort of their homes.

Our primary audience includes busy individuals, families, and modern consumers who value efficiency, transparency, and real-time access to product information. Customers can enjoy detailed product descriptions, live stock availability, customer reviews, and personalized account features that make shopping intuitive and stress-free.

What sets us apart is our powerful and comprehensive system. The platform offers real-time inventory validation, order status tracking, advanced search and filtering tools, and a fully featured administrative dashboard. Administrators can easily manage products, monitor stock levels, review sales analytics, and moderate customer feedback.

The application is built using JSP, backed by a SQL Server database, and deployed through Apache Tomcat in a containerized Docker environment ensuring flexibility, reliability, and smooth deployment to cloud services.

We want to create a loyal community who is tired of shopping with traditional sights and want to explore the sweeter and darker things in life.

SYSTEM FEATURES: Core Customer Features

1. Product Browsing and Search
Main product catalog displays all available items with images, prices, and stock levels
Multi-criteria search: filter by product name, category, and price range
Sorting options: name (A-Z, Z-A) and price (low to high, high to low)
Real-time stock availability indicators with color-coding (green for in-stock, yellow for limited, orange for low, red for out-of-stock)
Product rating display with star ratings and review counts

2. Product Detail Pages
Comprehensive product information including ID, category, price, and description
Large product image display with fallback support for multiple image formats
Average customer rating with stars and review count
Stock availability status with detailed unit counts
Disabled "Add to Cart" button for out-of-stock items

3. Shopping Cart Management
Add products to cart with automatic quantity tracking
View cart contents with product details and running total
Update item quantities with real-time validation
Remove items from cart
Stock validation prevents adding out-of-stock or insufficient inventory
Error messages for invalid cart operations

4. Customer Account Management
Create new customer account with data validation
Login/logout functionality with session management
View customer profile information (name, email, phone, address)
Edit profile page to update personal information
Change password functionality with current password verification
Password confirmation and minimum length validation

5. Order Management
Secure checkout process with shipping information
Order history page showing all previous orders with date filtering
Order details include product list, quantities, prices, and totals
Order status tracking (Pending, Processing, Shipped, Delivered, Cancelled)
Color-coded status badges for visual clarity

6. Product Reviews
View customer reviews on product detail pages
Average rating calculation and display
Star rating visualization (★★★★☆)

7. Admin Dashboard
Centralized navigation to all administrative functions
Daily sales chart with visual graph (Chart.js integration)
Sales report table by date
Quick access to customer management, products, orders, inventory, analytics, and reviews

8. Customer Management
List all customers with search functionality
View customer details including contact information and order history

9. Product Management
Add new products with name, description, price, category, and image
Edit existing product information
Delete products from catalog
Upload product images (supports multiple formats: .jpg, .webp, .avif, .png, .gif)

10. Inventory Management
View inventory levels across all warehouses
Inline editing of stock quantities
Update inventory with real-time validation
Low stock alerts dashboard with three severity levels:
Out of Stock (0 units) - Red alert
Critical (≤10 units) - Orange alert
Warning (11-20 units) - Yellow alert
Statistics cards showing counts for each alert level
Quick links to manage inventory and products

11. Order Management
View all orders with comprehensive filtering:
Date range filtering (start and end date)
Order amount filtering (min and max)
Update order status from admin interface
Five status levels: Pending, Processing, Shipped, Delivered, Cancelled
Process shipments with automatic inventory deduction
Generate tracking numbers (format: BBG-orderID-shipmentID-timestamp)

12. Sales Analytics Dashboard
Key performance statistics:
Total orders count
Total revenue
Average order value
Total customer count
Daily sales report for last 30 days with date and total

Input validation on forms
WEB AND AI SOURCES USED
AI Tools - Claude.ai/ Github Copilot
CSS Styling (all pages + Snake game styling, game logic from previous project)
Documentation (This document and most of the comments)

Oracle JSP Documentation - JavaServer Pages syntax and best practices
URL: https://docs.oracle.com/javaee/5/tutorial/doc/bnagx.html

Microsoft SQL Server Documentation - T-SQL syntax and functions
URL: https://learn.microsoft.com/en-us/sql/sql-server/

Chart.js Documentation - JavaScript charting library for sales graphs
URL: https://www.chartjs.org/docs/

Bootstrap Documentation - CSS framework for responsive design
URL: https://getbootstrap.com/docs/

Docker Documentation - Container configuration and deployment
URL: https://docs.docker.com/

Apache Tomcat Documentation - Application server configuration
URL: https://tomcat.apache.org/tomcat-9.0-doc/

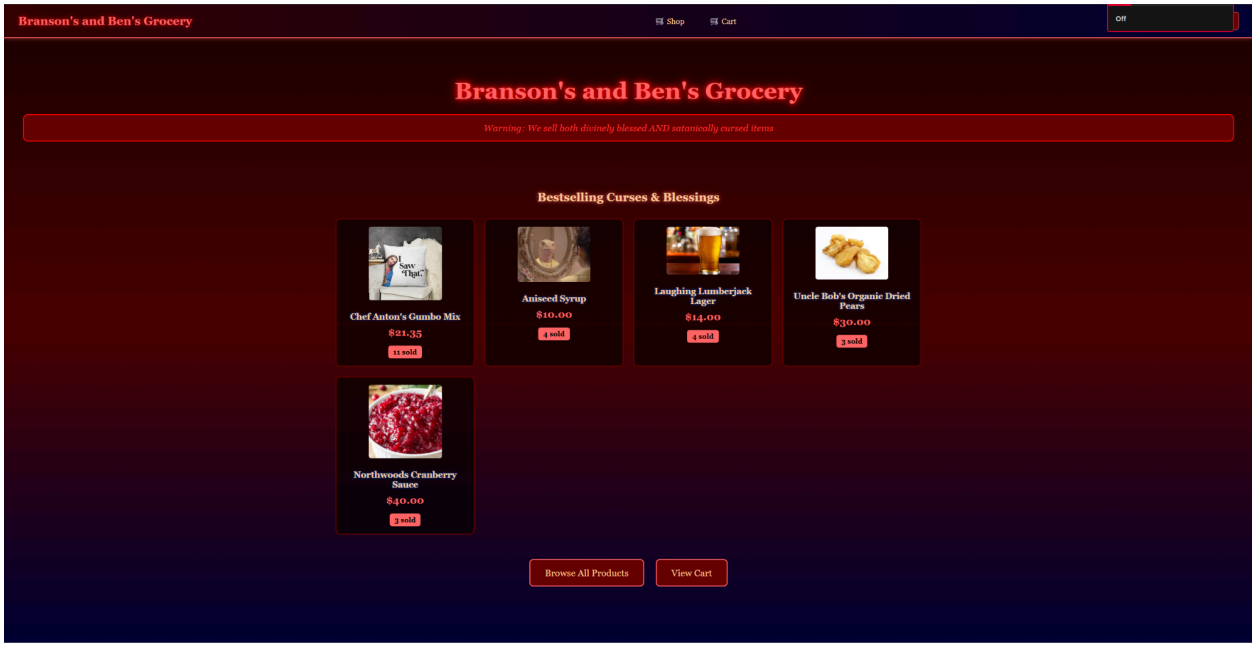
Learning Resources
W3Schools - HTML, CSS, and JavaScript reference
URL: https://www.w3schools.com/

Stack Overflow - Community solutions for specific technical challenges
URL: https://stackoverflow.com/

WALKTHROUGH WITH SCREENSHOTS

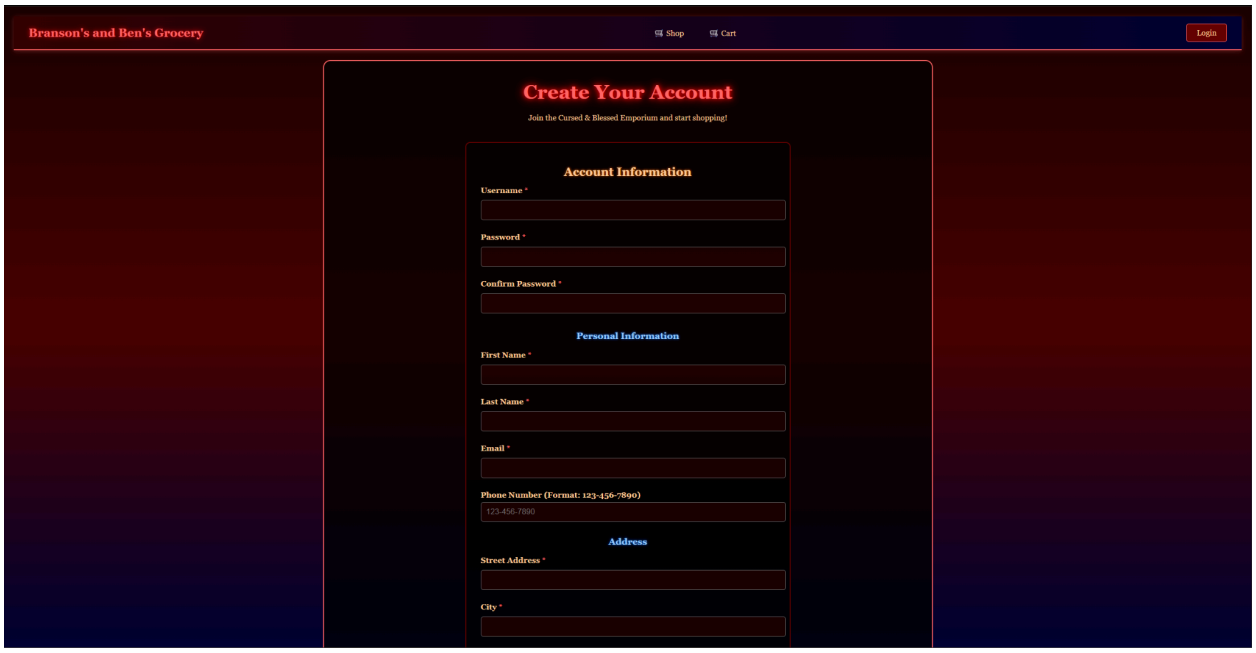
As A USER

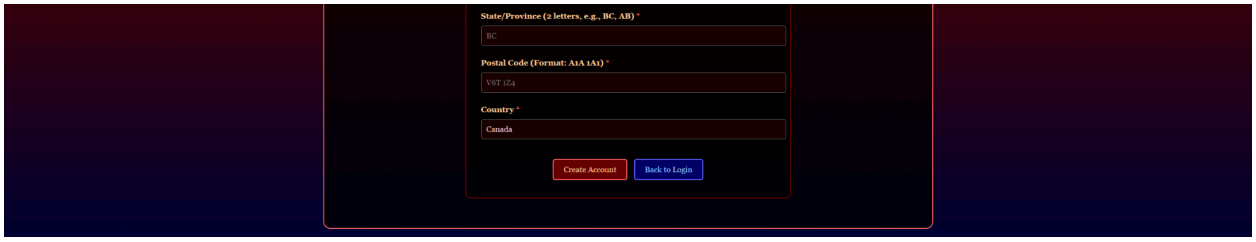
1. **Landing Page**



When a user first enters the page they're met with our home page, which displays our most sold items, they can begin shopping and browse all products, they can view their cart or log in.

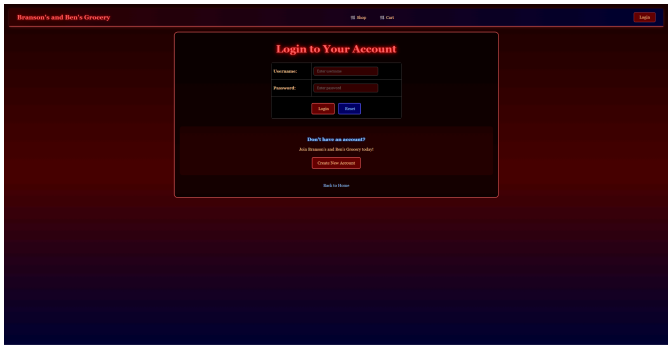
2. Create an account

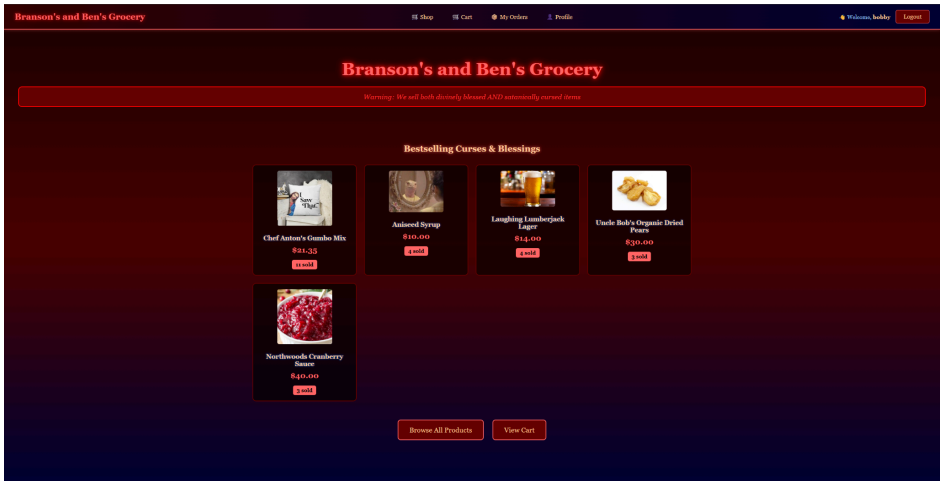




Next for our create account page, basic information will be asked for an account. Here we check if a username or email are already in use, as well as checking if the email is in the correct format. If they do try to use a username already in use an error message will appear telling them to choose another. Email will also show an error that email is already in use. As well checks if the passwords match

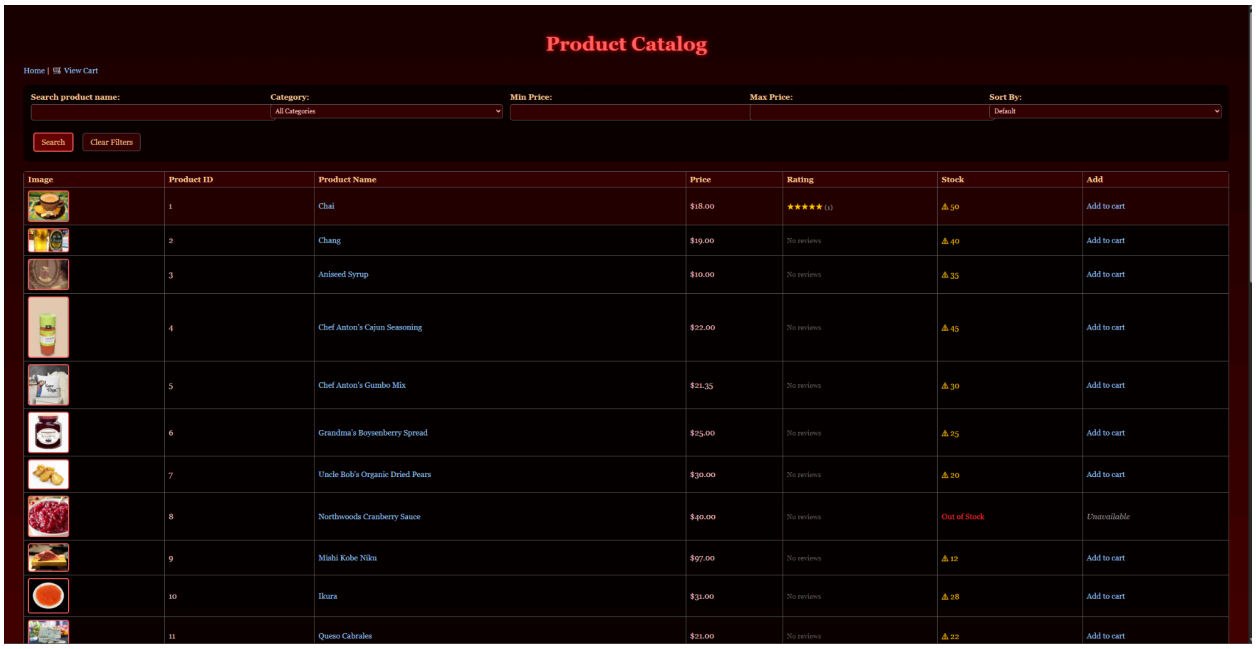
3. Login and session





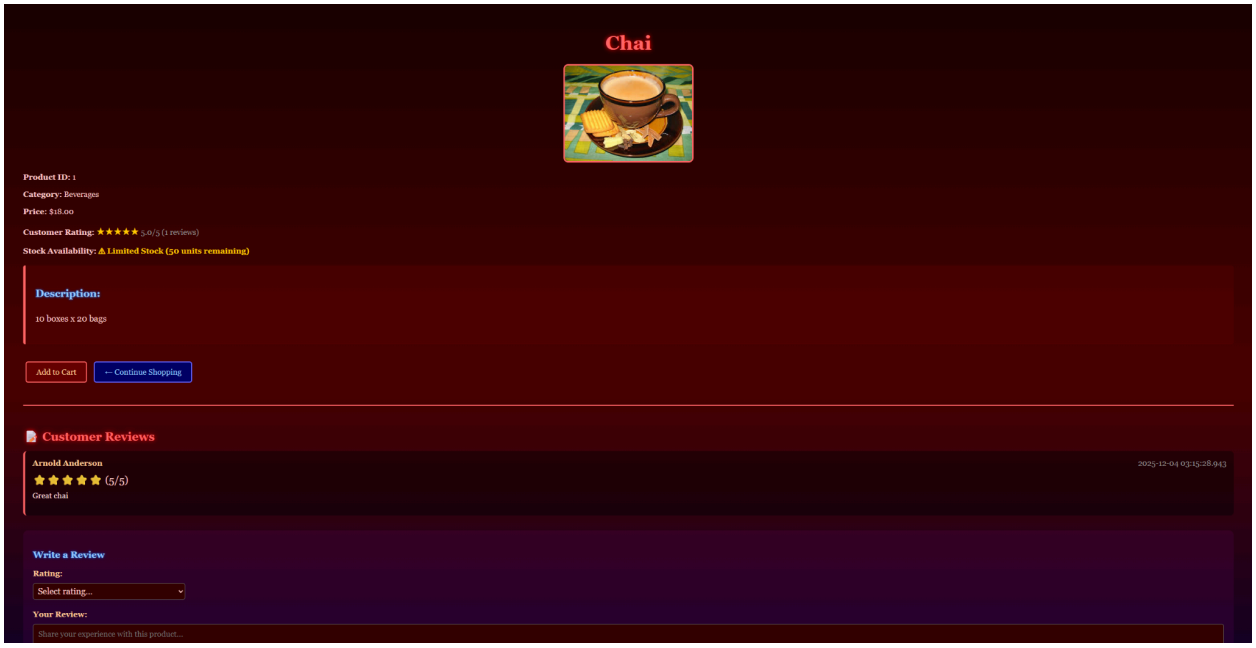
After a user logins, a message appears at the top of the screen welcoming them to the page. As well the nav bar now shows a My Orders, and Profile Button

4. Shopping Experience



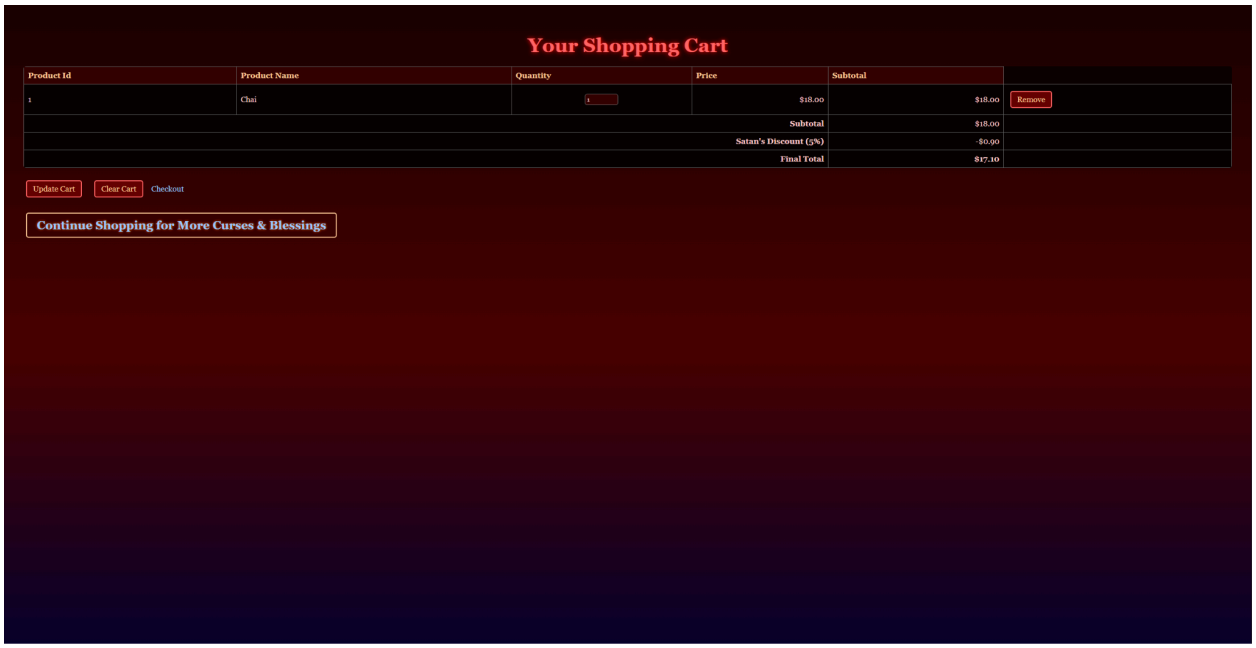
Here users are able to view all of our products, add them to their cart, view current inventory for them, as well search product by name, select a min or max price, and finally can sort products alphabetically and by price. As well they can filter by category.

5. Product Page



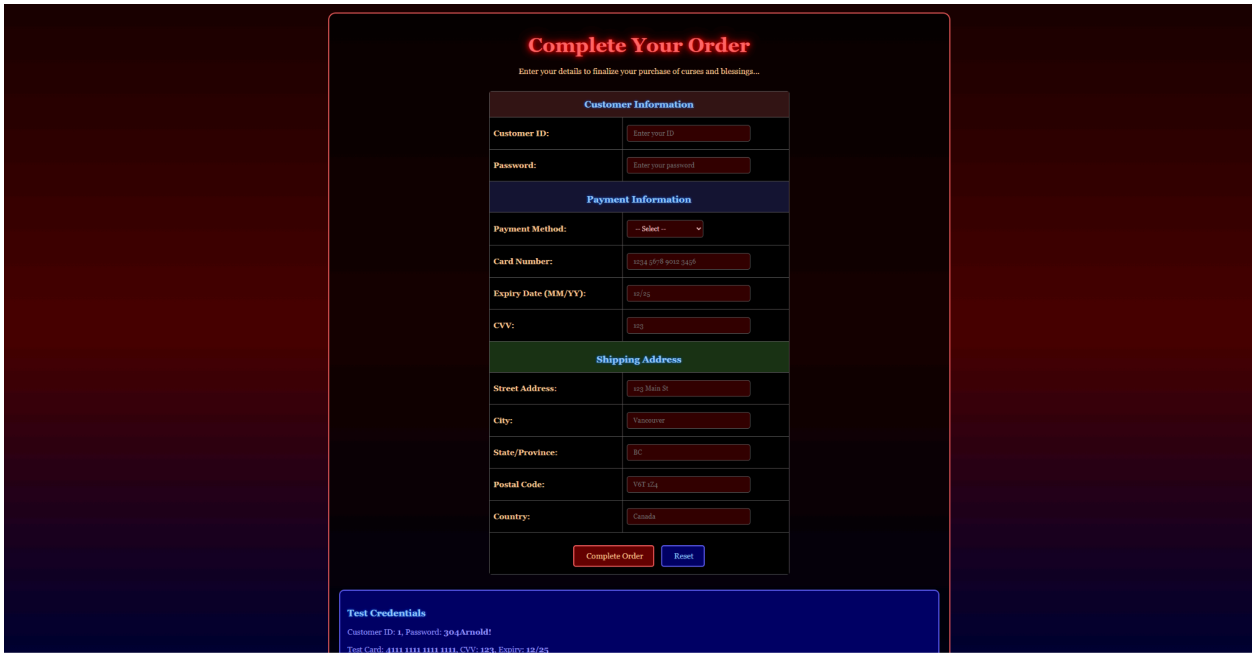
In the product page they get a more detailed description of the product and can as well look at reviews for that product. Users can only write reviews if they are logged into an account. They can add to their cart or return to browse other products.

6. Add to cart



The add to cart button will redirect the user to their shopping cart. Here users can update their cart by removing an item altogether or edit the quantity of items. From here they can continue shopping or check out.

7. Checkout

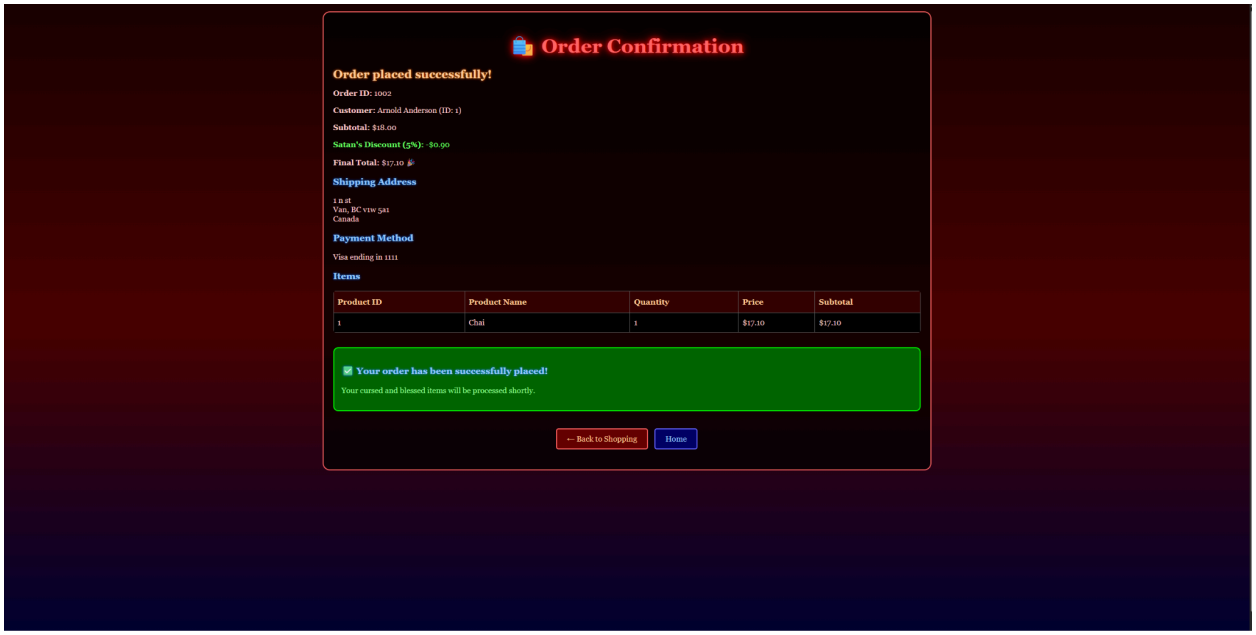


Here users are prompted to enter all of their information, and (fake) payment information, and shipping information. As well theres an option to play snake for a discount

8. Snake Game

If you win you get 5% off and if you lose your cart total is up 50%, discount or penalty is applied automatically.

9. Order Confirmation

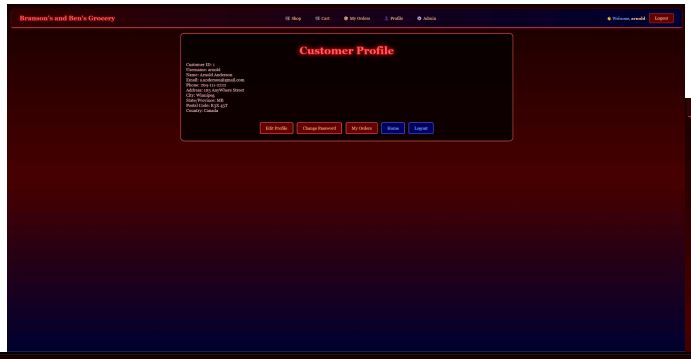


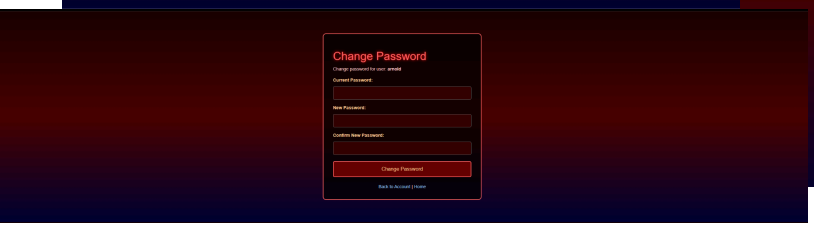
Displays the order confirmation and order details, as well gives you the option to return home or continue shopping

10. Order history

Displays all previous orders, the order status, as well allows you to search for specific orders by date.

11. Customer Profile page

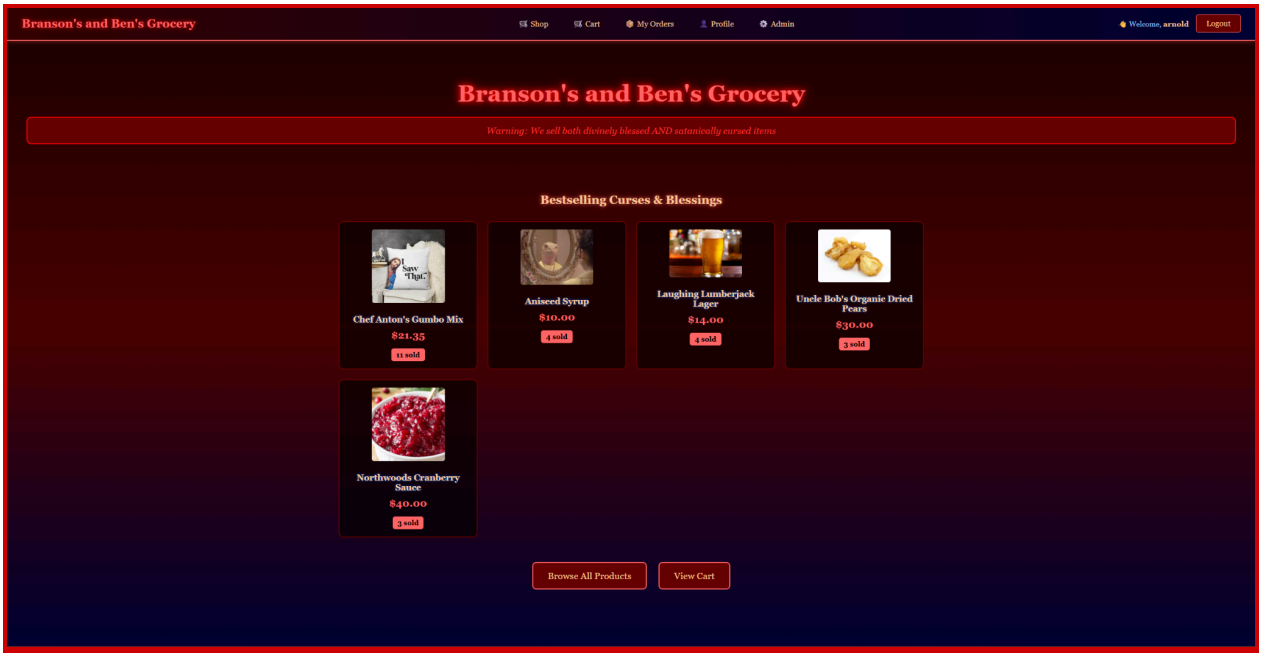




Here the user has many options, if they click on the edit your profile option they have the option to edit any of their basic information. Or they can change their password as well, which requires them to enter their old password and their two new ones.

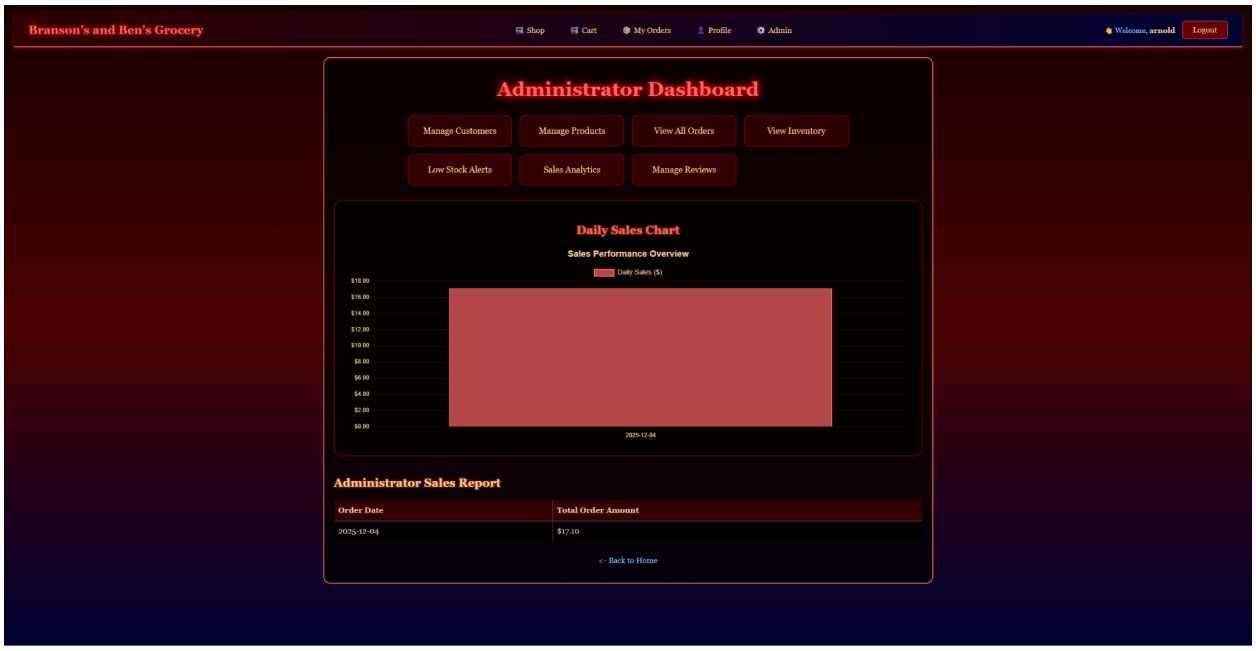
As A ADMIN

1. Home Page



The Custom User welcome message appears as normal, but it also shows them the Admin option in the Nav Bar.

2. Admin Dashboard



The admin homepage shows the sales total for the day.

3. The Inventory Page

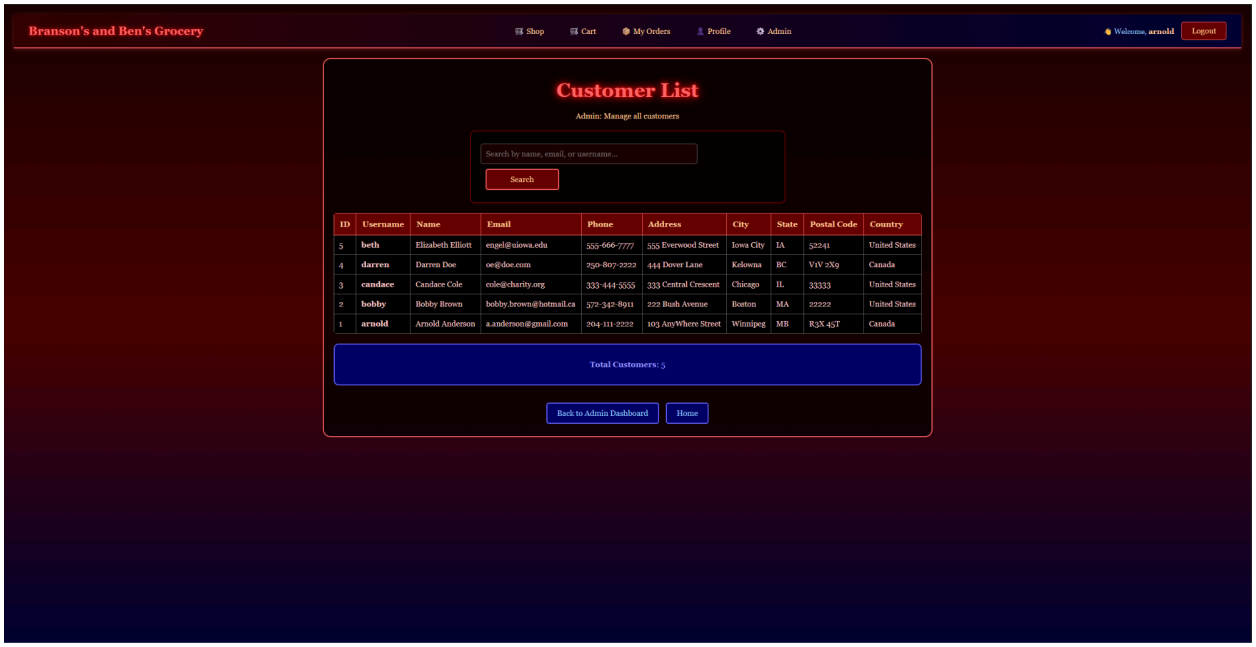
Product ID	Product Name	Category	Quantity In Stock	Price	Total Value	Status	Action
1	Chai	Beverages	50 Update	$18.00	$900.00	In Stock	
2	Chang	Beverages	40 Update	$19.00	$760.00	In Stock	
3	Aniseed Syrup	Condiments	35 Update	$10.00	$350.00	In Stock	
4	Chef Anton's Cajun Seasoning	Condiments	46 Update	$22.00	$1,012.00	In Stock	
5	Chef Anton's Gumbo Mix	Condiments	30 Update	$21.35	$640.50	In Stock	
6	Grandma's Boysenberry Spread	Condiments	25 Update	$25.00	$625.00	In Stock	
7	Uncle Bob's Organic Dried Pears	Produce	20 Update	$30.00	$600.00	In Stock	
8	Northwoods Cranberry Sauce	Condiments	0 Update	$40.00	$0.00	OUT OF STOCK	
9	Mishi Kobe Niku	Meat/Poultry	12 Update	$97.00	$1,164.00	In Stock	
10	Ikura	Seafood	28 Update	$31.00	$868.00	In Stock	
11	Queso Cabrales	Dairy Products	22 Update	$21.00	$462.00	In Stock	
12	Queso Manchego La Pastora	Dairy Products	35 Update	$38.00	$1,330.00	In Stock	
13	Tofu	Produce	40 Update	$6.00	$240.00	In Stock	
14	Genen Shouyu	Condiments	32 Update	$23.25	$744.00	In Stock	
15	Pavlova	Confections	18 Update	$17.45	$314.10	In Stock	
16	Alice Mutton	Meat/Poultry	25 Update	$39.00	$975.00	In Stock	
17	Carnarvon Tigers	Seafood	30 Update	$62.50	$1,875.00	In Stock	
18	Teatime Chocolate Biscuits	Confections	45 Update	$9.20	$414.00	In Stock	
19	Sir Rodney's Marmalade	Confections	28 Update	$81.00	$2,268.00	In Stock	
20	Sir Rodney's Scones	Confections	33 Update	$10.00	$330.00	In Stock	
21	Gustaf's Knackebread	Grains/Cereals	50 Update	$21.00	$1,050.00	In Stock	
22	Sasquatch Ale	Beverages	38 Update	$14.00	$532.00	In Stock	
23	Steeleye Stout	Beverages	42 Update	$18.00	$756.00	In Stock	
24	Inlagd Sill	Seafood	27 Update	$19.00	$513.00	In Stock	
25	Boston Crab Meat	Seafood	20 Update	$18.40	$368.00	In Stock	
26	Jack's New England Clam Chowder	Seafood	35 Update	$9.65	$337.75	In Stock	
27	Singaporean Hokkien Fried Mee	Grains/Cereals	24 Update	$14.00	$336.00	In Stock	
28	Louisiana Fiery Hot Pepper Sauce	Condiments	55 Update	$21.05	$1,157.75	In Stock	
29	Laughing Lumberjack Lager	Beverages	48 Update	$14.00	$672.00	In Stock	
				Warehouse Total Value:	$21,594.10		

Total Products: 29 | Out of Stock: 1 | Low Stock: 0 | In Stock: 28

Back to Admin Dashboard <- Home

Here it shows every product in that warehouse as well as the price and total value of each product and the entire warehouse. As well we have the option to update the quantity for each product and a green bar will appear if they successfully update.

4. Customer List



ID	Username	Name	Email	Phone	Address	City	State	Postal Code	Country
5	beth	Elizabeth Elliott	engel@uiowa.edu	555-666-7777	555 Everwood Street	Iowa City	IA	52241	United States
4	darren	Darren Doe	oe@doe.com	250-807-2222	444 Dover Lane	Kelowna	BC	V1V 2X9	Canada
3	candace	Candace Cole	cole@charity.org	333-444-5555	333 Central Crescent	Chicago	IL	33333	United States
2	bobby	Bobby Brown	bobby.brown@hotmail.ca	572-342-8911	222 Bush Avenue	Boston	MA	22222	United States
1	arnold	Arnold Anderson	a.anderson@gmail.com	204-111-2222	103 AnyWhere Street	Winnipeg	MB	R3X 45T	Canada

In this page we have the ability to view all accounts that are registered and we have the ability search for customers.

5. Search Orders

Order Id	Order Date	Customer Id	Customer Name	Total Amount	Status	Action
1002	2025-12-04 04:51:49.443	1	Arnold Anderson	$17.10	Pending	Pending Update

Product Id	Quantity	Price
1	1	$18.00

5	2019-10-15 10:25:55.0	5	Elizabeth Elliott	$277.40	Pending	Pending Update

Product Id	Quantity	Price
5	4	$21.35
19	2	$81.00
20	3	$10.00

4	2019-10-17 05:45:11.0	2	Bobby Brown	$327.85	Pending	Pending Update

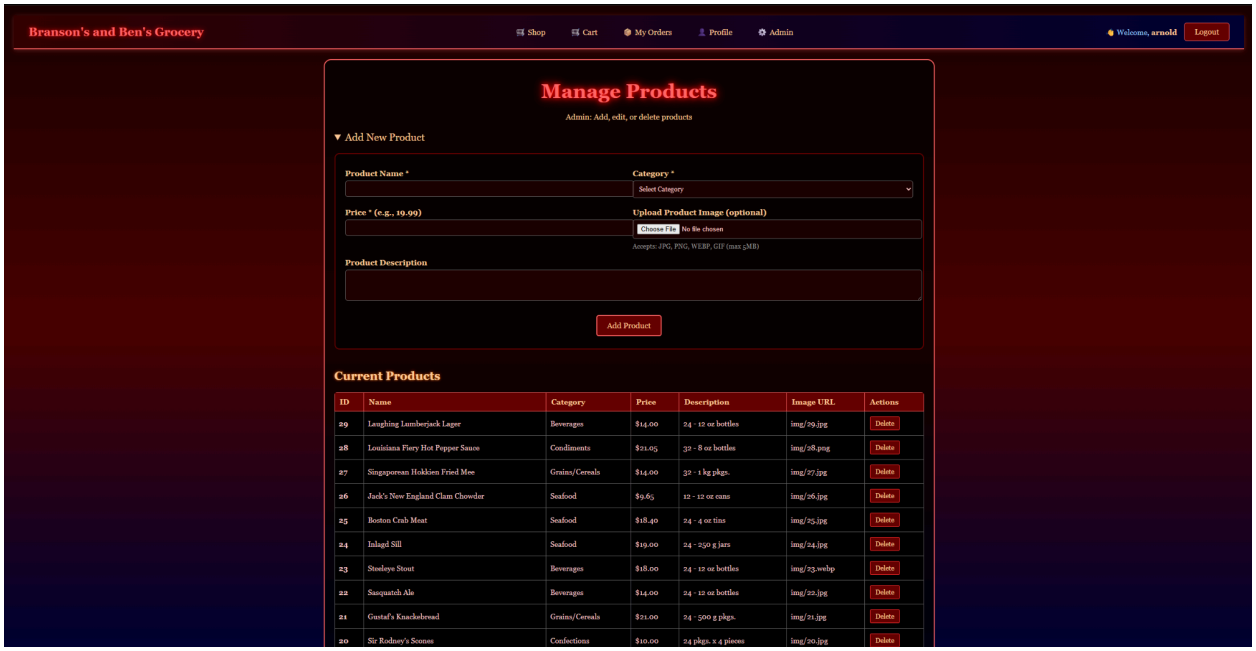
Product Id	Quantity	Price
3	4	$10.00
8	3	$40.00
13	3	$23.25
28	2	$21.05
29	4	$14.00

3	2019-10-15 03:30:22.0	3	Candace Cole	$140.00	Pending	Pending Update

Product Id	Quantity	Price
6	2	$25.00

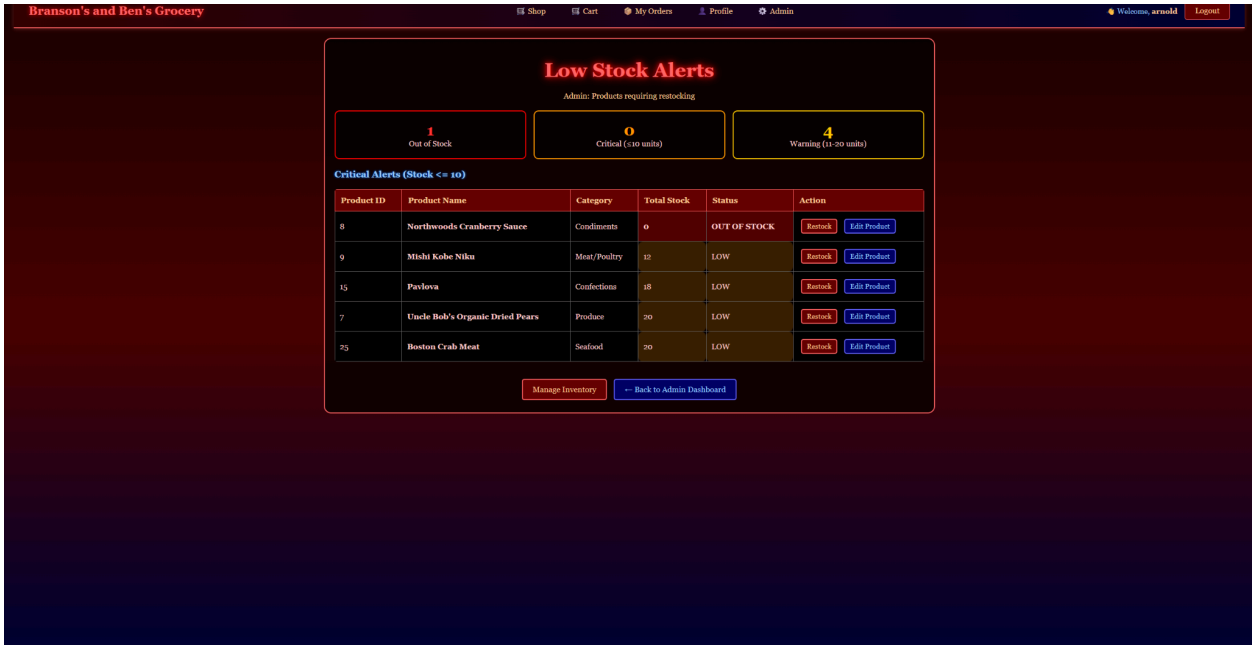
Admin can see all the orders in database and can edit their status from pending, processing, shipping, delivered, and cancelled

6. Manage products



Here we can view all products, add new products, or delete products. The add product form will validate all data in the form before sending the request.

7. Low Stock Alert



Page that shows products that are low in stock or out of stock, gives the admin the option to restock the product.

8. Sales Analytics

Sales Analytics Dashboard

6	$960.80	$160.13	5
Total Orders	Total Revenue	Average Order Value	Total Customers

Daily Sales (Last 30 Days)

Date	Orders	Total Sales
2025-12-04	1	$17.10

Top 10 Products by Revenue

Rank	Product	Units Sold	Total Revenue
1	Chef Anton's Gumbo Mix	11	$234.85
2	Sir Rodney's Marmalade	2	$162.00
3	Northwoods Cranberry Sauce	3	$120.00
4	Uncle Bob's Organic Dried Pears	3	$90.00
5	Tofu	3	$69.75
6	Laughing Lumberjack Lager	4	$56.00
7	Grandma's Boysenberry Spread	2	$50.00
8	Louisiana Fiery Hot Pepper Sauce	2	$42.10
9	Aniseed Syrup	4	$40.00
10	Chai	2	$36.00

Top 10 Customers by Total Spending

Rank	Customer	Email	Orders	Total Spent
1	Bobby Brown	bobby.brown@hotmail.ca	2	$434.60
2	Elizabeth Elliott	engel@uiowa.edu	1	$277.40
3	Candace Cole	cole@charity.org	1	$140.00
4	Arnold Anderson	a.anderson@gmail.com	2	$108.80

Sales by Category

Category	Units Sold	Total Revenue
Condiments	22	$486.95
Confections	5	$192.00
Produce	6	$159.75
Beverages	6	$92.00
Seafood	1	$31.00

Here it shows the total orders, the orders in the last month, the best sellers, the customers that spend the most as well as the best selling categories.

CATEGORY / FEATURE	**TYPE**	**POSS**	**EARN**	**COMMENTS**
CATEGORY / FEATURE	TYPE	POSS	EARN	COMMENTS / CODE FILES
DOCUMENTATION (15 pts)				
Executive Summary	Required	2	2	Complete executive summary provided
System Features (2 pages)	Required	2	2	Comprehensive features listed
List of Web/AI sources	Required	3	3	AI tools, Bootstrap, Chart.js
Walkthrough w/ screenshots	Required	8	8	27 screenshots, detailed
HOSTING (10 pts)				
Running on cloud server	Advanced	10	0	Runs locally via Docker
MAIN / HOME PAGE (14 pts)				
Search products by name	Core	1	1	listprod.jsp
Browse by category	Core	1	1	listprod.jsp
List products	Core	1	1	listprod.jsp
List products w/ images	Core	1	1	listprod.jsp
Header menu	Basic	1	1	header.jsp
Header shows logged-in user	Basic	1	1	header.jsp
Dynamic homepage products	Basic	2	0	Not implemented
Improved UI	Basic	2	2	CSS, gradients, star ratings
Product recommendation	Advanced	4	0	Not implemented
SHOPPING CART (8 pts)				
Add to cart	Core	1	1	addcart.jsp
View cart	Core	1	1	showcart.jsp
Update quantity	Basic	1	1	updatecart.jsp
Remove item	Basic	1	1	updatecart.jsp
Improved formatting/UI	Basic	1	1	Custom styling
Cart stored in database	Basic	3	3	cartDB.jsp
CHECKOUT (9 pts)				
Checkout w/ customer ID	Core	1	1	order.jsp
Checkout w/ payment & shipping info	Basic	2	2	checkout.jsp
Checkout validation	Basic	2	2	order.jsp
Multiple shipments	Advanced	2	0	Not implemented
Tax/shipping calc	Advanced	2	0	Not implemented
PRODUCT DETAIL PAGE (2 pts)				
Product detail page	Core	1	1	product.jsp

Product image	Core	1	1	product.jsp
USER ACCOUNTS / LOGIN (12 pts)				
Create account page	Basic	2	2	register.jsp
Account validation	Basic	2	2	registerAction.jsp
Edit account info	Basic	2	2	customer.jsp, changePassword.jsp
Login/logout	Core	1	1	validateLogin.jsp
Forgot password email	Advanced	4	0	Not implemented
Order history page	Basic	1	1	myorders.jsp
PRODUCT REVIEWS (5 pts)				
Enter a review	Basic	2	2	product.jsp, submitReview.jsp
Display reviews	Basic	1	1	product.jsp
Limit 1 review per user	Advanced	2	0	Not implemented
WAREHOUSE / INVENTORY (5 pts)				
Display inventory	Basic	2	2	inventory.jsp
Edit inventory	Advanced	3	3	inventory.jsp, updateStock.jsp
ADMIN PORTAL (18 pts)				
Secured by login	Core	1	1	auth.jsp
List all customers	Basic	1	1	listcust.jsp
Sales/orders report	Basic	1	1	listorder.jsp
Graph report	Advanced	3	3	Chart.js
Add product	Basic	2	2	addProduct.jsp
Update/delete product	Basic	2	2	manageProducts.jsp
Ship / update order status	Basic	1	1	ship.jsp
Upload photo to filesystem	Advanced	2	0	Not implemented
Upload photo to database	Advanced	4	4	AJAX upload
SQL database restore	Core	1	1	loaddata.jsp
Add/update warehouse/customer	Basic	2	0	Warehouse only
DATABASE / GENERAL				
DB triggers validation	Advanced	2	0	Not implemented
JavaScript validation	Advanced	4	4	register.jsp
UI / usability	—	5	5	Responsive design, gradients
Additional features	—	5	5	Review moderation, low stock alerts, filters, tracking numbers, star ratings
TOTAL POINTS		60	60	Max 50 + 10 bonus